

DARSINI LAKSHMIAH

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Marketing teams at AARP used to file tickets and wait days for data answers. Then I built an RAG-based NL-to-SQL assistant on 76M+ behavioral records and they stopped. Before that: AML fraud detection on 1.5M+ daily banking transactions at NatWest, and a \$200M grocery analytics platform at Amazon. Four years of production ML across banking, retail, marketing.

WORK EXPERIENCE

AARP SERVICES INCORPORATED, Washington, DC, USA

Data Scientist Intern

September 2025 – March 2026

- **Cut analytics turnaround 50%** by shipping a production RAG NL-to-SQL assistant (LangChain, ChromaDB, Databricks) with schema-aware prompt engineering and query validation - marketing teams moved from filing data tickets to self-serving campaign insights in natural language within two weeks of launch.
- **Lifted campaign engagement ~10%** by building a LightGBM recommendation system on **76M+** behavioral records - measured via CTR, content consumption rate, and downstream engagement metrics compared against baseline approaches; tracked model versions, feature importance, and precision/recall with MLflow.
- Drove self-service analytics adoption - trained marketing and CRM teams on the NL-to-SQL assistant, significantly reducing recurring analyst data requests within the first month of deployment.
- Standardized the team's data foundation by authoring dbt pipelines on Databricks with lineage tracking, quality tests, and governed transformations - eliminating ad-hoc query inconsistencies across marketing and CRM reporting.

AMAZON, Austin, Texas, USA

Business Intelligence Engineer Intern

May 2025 - August 2025

- Built a customer 360 analytics platform for a **\$200M** Amazon Fresh + Whole Foods initiative, unifying **87M+** transactions across **530+** stores to surface lifetime value, purchasing patterns, and high-value customer segments; product managers used the dashboard to evaluate Dash Cart and Click-Checkout adoption patterns and prioritize retail experimentation.
- Developed a purchase-propensity model - engineered features on purchase frequency, basket size, and browsing history; validated model performance and integrated predictions into retail dashboards guiding feature launch decisions.
- Influenced Amazon Fresh + Whole Foods in-store strategy by presenting store-level Dash Cart and Click-Checkout adoption analysis to PMs and retail leadership - findings directly shaped the integrated rollout roadmap.
- Reduced dashboard latency by **20%** by rebuilding ETL pipelines to transform raw transactions into optimized aggregation layers for scalable retail analytics.

NATWEST GROUP, Chennai, India

Senior Data Analyst

Aug 2021 - June 2024

- Improved query performance **35%** - engineered ETL pipelines (Azure Data Factory, Snowflake, ADLS Gen2) processing 1.5M+ daily banking transactions across card payments, ACH, and digital activity streams; optimized Snowflake schemas and KPI layers for product, risk, and compliance teams.
- Strengthened AML monitoring - built NLP anomaly detection (TF-IDF + XGBoost) on **1.5M+** daily transactions; partnered with compliance investigators, performed feature selection and hyperparameter tuning, delivered explainable risk scoring with feature importance reports and Power BI dashboards for investigator prioritization.
- Drove **9%** campaign uplift - built customer segmentation models and A/B experiments (Python, Snowflake, statsmodels); translated clustering outputs into recommendations on segment response rates, revenue opportunity, and compliance considerations - influencing leadership to adopt personalized targeting strategies.

EDUCATION

George Washington University, Washington, DC - Master of Science in Data Science, GPA - 3.97/4.0

May 2026

Madras Christian College, Chennai, India - Bachelor of Science in Mathematics, CGPA - 8.50/10

May 2021

PROJECTS

AI Interview Coach | [GitHub](#) | [Website](#) | AutoGen · GPT-4o · FastAPI · Whisper · WebSocket · face-api.js

- Built a real-time multi-agent interview simulation where AI interviewer, evaluator, and candidate agents conduct company-specific mock interviews over WebSocket with live Whisper transcription and webcam-based performance scoring.

NutriNav Food Access Intelligence App | [GitHub](#) | [DevPost](#) | XGBoost · Groq Whisper · FastAPI · Supabase

- Top 3 of 35 teams, George Hacks × UN Reboot the Earth. Built a multilingual voice-first app connecting 75,000+ DC SNAP residents to real-time corner store inventory; Whisper auto-detects language, LLaMA 3.3 70B responds in kind, and XGBoost grades products A - D via USDA Data

TECHNICAL SKILLS

Programming Languages: Python, R, SQL, SAS, PySpark

ML & AI: Scikit-learn, PyTorch, MLflow, LangChain, LangGraph, AutoGen, LlamaIndex, RAG, MCP, Prompt Engineering

Cloud & Data Engineering: AWS (Glue, Redshift, Athena, S3, Bedrock), Azure Data Factory, ADLS Gen2, dbt, Airflow

Databases & Vector Stores: Databricks, Snowflake, MySQL, MongoDB, Neo4j, FAISS, ChromaDB

Analytics & Modeling: A/B Testing, Causal Inference, NLP, Time Series Forecasting, Recommendation Systems, Behavioral Analytics, KPI Reporting, Data Modeling

Visualization & Tools: Tableau, Power BI, Looker, QuickSight, Pandas, NumPy, Excel (Advanced), Git/GitHub, Docker, Flask